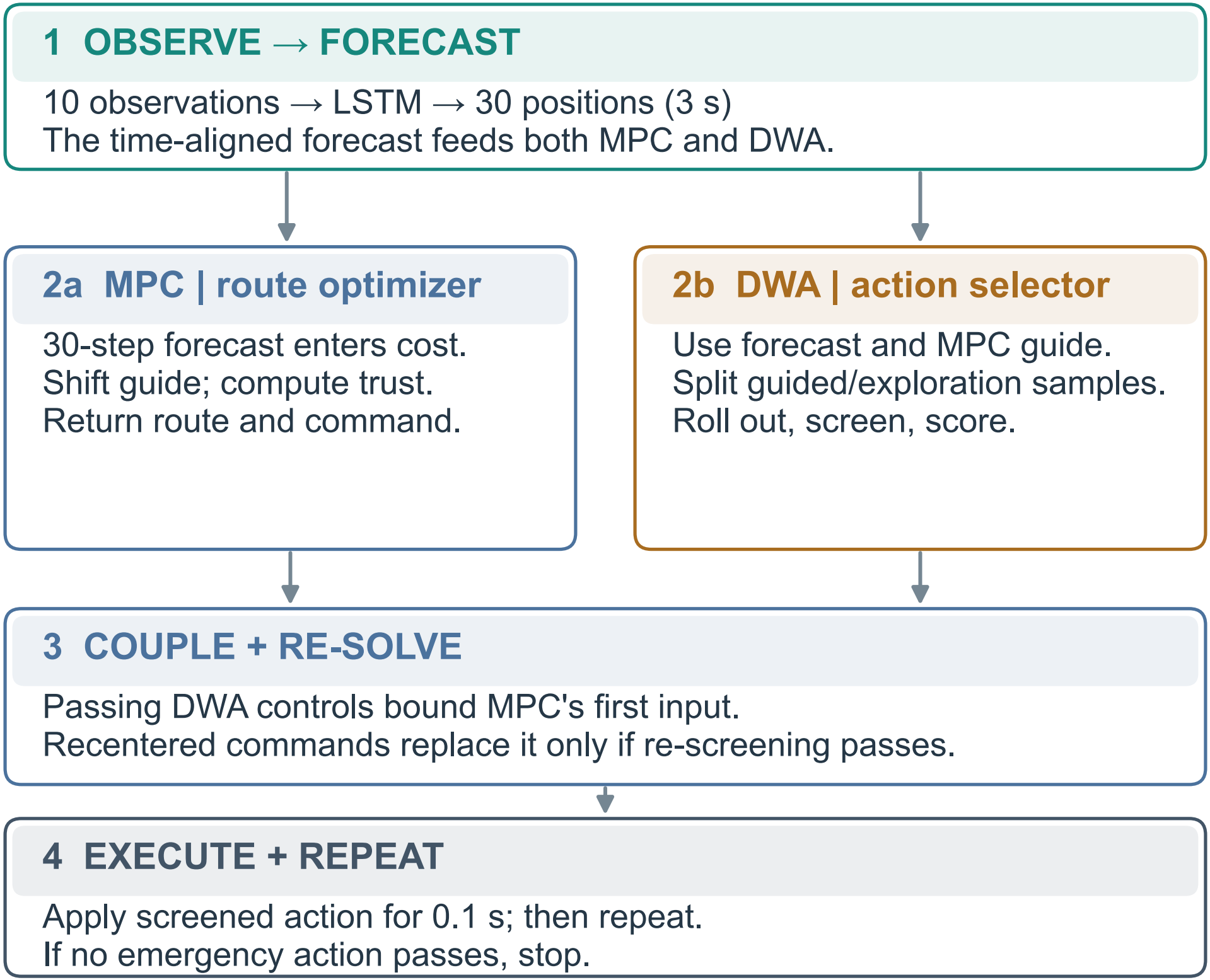
1 OBSERVE → FORECAST
10 observations → LSTM → 30 positions (3 s)
The time-aligned forecast feeds both MPC and DWA.
2a MPC | route optimizer
30-step forecast enters cost.
Shift guide; compute trust.
Return route and command.
2b DWA | action selector
Use forecast and MPC guide.
Split guided/exploration samples.
Roll out, screen, score.
3 COUPLE + RE-SOLVE
Passing DWA controls bound MPC's first input.
Recentered commands replace it only if re-screening passes.
4 EXECUTE + REPEAT
Apply screened action for 0.1 s; then repeat.
If no emergency action passes, stop.